

Objective

To harness my extensive mechanical engineering background and 5 years of software development expertise alongside my experience in Master's in Robotics to drive groundbreaking advancements in the field of autonomous robotics, especially in the planning sector.

Education

- **Master of Science in Engineering Science: Robotics, University at Buffalo, The State University of New York, Expected: June 2025**
- **Bachelor of Technology: Mechanical Engineering, SASTRA Deemed University, (GPA 7.96/10), May 2018**

Technical Skills

- **Programming Languages:** Java, C, C++, Python, Go, HTML, SQL, JavaScript, Java Server Pages(JSP)
- **Tools and Frameworks:** REST, Agile, Waterfall, Microsoft Word, Excel, PowerPoint, Outlook, Linux, Git, Mercurial, MySQL, PostgreSQL, Kafka, Docker, gRPC, Protobuf, Apache Lucene, Elasticsearch, Github, Gitlab, Bitbucket, Sourcetree, SolidWorks, CATIA, Pro-E, ANSYS, LS-Dyna, ROS, rospy, Simulink, MATLAB, LaTeX

Experience

Research Assistant at DRONES Lab, SUNY Buffalo, NY

May

2024-Present

Ongoing Research – Safe Planning of Robotic Manipulators: Developed **safe (collision-free) trajectory generation** for a **2-DoF planar arm** using **Control Barrier Functions (CBFs)** that incorporate arm kinematics.

- Implemented **quadratic programming and optimization** using **CasADi** for trajectory planning.
- Currently extending to **3D and multi-DoF arms**, exploring **generative methods** for safe trajectory generation based on the CBF safe plan.

Member Technical Staff (Software Development Engineer), Zoho Corporation, Chennai, Tamil Nadu, India

Nov 2016- Dec 2016

- Spearheaded core development of the Cloud Repository Management System, overseeing end-to-end service creation from competitor analysis to feature implementation, server provisioning, and security enhancement.
- Engineered search indexing system achieving sub-40 ms retrieval time, optimizing content search functionality.
- Implemented dynamic code intelligence across multiple programming languages within repositories using docker containers, reducing redundancy and enhancing efficiency.
- Innovated database architecture reducing data redundancy by over 70% and nullifying server crashes through background process optimization, automated heavy-duty repository processes, reducing response time by 60% through asynchronous pipeline scheduling.
- Monitored and addressed security threats adhering to cloud software norms, ensuring seamless operations.
- Integrated polyglot services via gRPC and protobuf, reducing maximum response time by over 200%.
- Produced comprehensive product, API, and technical documentation fostering peer collaboration and conducted informative sessions for product induction and technical presentations.

Academic Projects

Deep Learning :

Jan 2024 - May

2024

- Implemented object detection using Vision Transformer (ViT) with Patch Embeddings in PyTorch on the Caltech-101 airplanes dataset, achieving a mean IoU of 88.4%

Computer Vision :

Aug 2024 - Dec

2024

- Implemented **PointNet** in **PyTorch** for **semantic segmentation of point clouds**, benchmarking against **MMDetection3D**
- Experimented with attention mechanisms in PointNet and MMDetection3D pipelines.
- Explored **loss function combinations** (correlation, focal, dice, Jaccard) to enhance performance.
- Trained on a **sequentially transformed SemanticKITTI dataset**, achieving **71% training accuracy** and **62% test accuracy** across variations.

SEAS Partnership Program - EARTH:

May 2024 -

Present

- Developed autonomous navigation and control for excavators and earth-movers, integrating perception, planning, and control for embodied autonomy. Led **software development, programming, and autonomous navigation stacks**
- Designed **global navigation** using **Nav2** with multiple planners and differential drive configurations.
- Implemented **arm trajectory planning** using **MovelIt**, configuring URDF-based planning groups and controllers.
- Developed **ROS2 lifecycle management**, synchronizing nodes, topics, and messages, and interfaced with **excavator hydraulics via CAN**.

Robotics Algorithms

Aug 2023 - Dec 2023

- Implemented **Rapidly-Exploring Random Trees**, **A***, and **Dijkstra** algorithms for path planning.
- Developed **Gap Follow** algorithm for autonomous vehicles and secured second position in the race organized as part of the program.
- Utilized **AMCL**, **Particle Clouds**, **Pure Pursuit** for **localization and control**.
- Researched **Stereo & Monocular Odometry** for **perception** and **motion estimation**.

Senior Undergraduate Year Project: Side Crash Analysis of a Tubular Frame

Jan 2018 - May 2018

- **Side Pole Impact Test** to determine the **crashworthiness of a tubular vehicle frame** using CATIA and LS-Dyna
- Proposed and **simulated the decrease in frame deformation** by introducing a woven flax-epoxy composite side structure that **absorbs part of the crash energy** thus protecting the passenger cabin
- Documentation of the impact parameters and the level of deformation and energy absorbed, with and without the composite side structure

SASTRA Racing Team

Jun 2015 - Dec 2016

Sugheerth Sreedharan

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Part of the **SASTRA SAE**(Society of Automotive Engineers) **Collegiate Club** and a member of the **SASTRA Racing Team(Transmission Team)**, building and testing an off-road buggy to compete in the endurance racing event: **BAJA SAEINDIA 2016**

Responsibilities And Activities:

- Research and selection of a suitable differential, calculation of output stats and computation of maximum gradability
- Creating digital models of the differential casing using SolidWorks, performance and analysis reports
- Fabrication, Building and testing the vehicle, tuning the engine for better performance and integrating the engine into the vehicle